

Subspaces, Linear Dependence, Basis and Dimension

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Outline

- 1 Vector Spaces and Subspaces
- 2 Linear Combinations and Span
- 3 Linear Dependence and Independence
- 4 Basis
- 5 Dimension
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Vector Spaces: Recall

A **vector space** V over a field $\mathbb{F}$ is a set equipped with:

- vector addition $+$,
- scalar multiplication.

subject to the standard axioms (already defined).

Examples:

✓ $\mathbb{R}^n$

- Space of polynomials $\mathcal{P}$
- $C[a, b]$ (the space of continuous functions)

$(V, \mathbb{R}) \leftarrow$



vector space

$\mathcal{P}_2$: the space of polyⁿ of order ≤ 2 .

$$p(t) = a_0 + a_1 t + a_2 t^2$$

$a_0, a_1, a_2 \in \mathbb{R}$

$$q(t) = b_0 + b_1 t + b_2 t^2$$

$$0 \in \mathcal{P}_2 \quad c \cdot p(t) = c p(t) = (a_0 c) + (a_1 c) t + (a_2 c) t^2 \quad p(t) + q(t) = (a_0 + b_0) + (a_1 + b_1) t + (a_2 + b_2) t^2$$

Example of a Vector Space

Example (Example of a Vector Space)

Let $C[a, b]$ be the set of all real-valued continuous functions on $[a, b]$. Let $f(x) + g(x) = (f + g)(x)$ and $\alpha f(x) = (\alpha f)(x)$ for $f, g \in C[a, b]$ and $\alpha \in \mathbb{R}$. Show that $C[a, b]$ is a vector space over $\mathbb{R}$.

$$\left. \begin{array}{l} f + g \in C[a, b] \\ \alpha f \in C[a, b] \end{array} \right\}$$

0 as a constant
is a cont^s fⁿ on $[a, b]$.
 f is cont^s $\Rightarrow -1 \cdot f$ is
also cont^s.

check!

$\Rightarrow (C[a, b], \mathbb{R})$ is a vector space. $f + (-f) = 0$.

Subspaces

Definition (Subspace)

A subset $W \subseteq V$ is called a *subspace* of V if W is itself a vector space under the same operations.



Subspace Test

vector zero
(additive
identity)

A nonempty subset $W \subseteq V$ is a subspace if:

- 1 $0 \in W$
- 2 $u, v \in W \Rightarrow u + v \in W$
- 3 $u \in W, \alpha \in \mathbb{F} \Rightarrow \alpha u \in W$

$\mathbb{R}$

$\mathbb{R}^n$

$$0_W = (0, 0, 0, \dots, 0)$$

n -many.

$\mathbb{P}_2$

$$0 = 0 + 0t + 0t^2$$

Examples of Subspaces

$(0,0,0) \in W$.
 $u_1 = (x_1, y_1, 0)$
 $u_2 = (x_2, y_2, 0)$
 $u_1 + u_2 = (x_1, y_1, 0) + (x_2, y_2, 0)$
 $= (x_1 + x_2, y_1 + y_2, 0)$
 $\in W$.

$\bullet \{(x, y, 0) : x, y \in \mathbb{R}\} \subset \mathbb{R}^3$

$\bullet$ Solution set of a homogeneous linear system

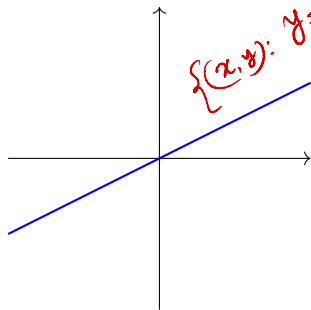
$c u_1 = c(x_1, y_1, 0)$
 $= (cx_1, cy_1, 0)$
 $\in W$.

Not a subspace:

$$\{(x, y, 1) : x, y \in \mathbb{R}\}$$

(does not contain zero vector)

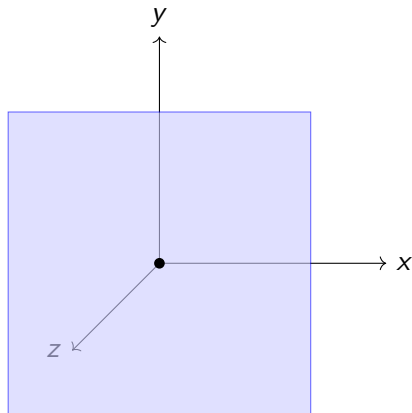
Subspace Example



$$\{(x, y) : y = mx\} \subseteq \mathbb{R}^2$$

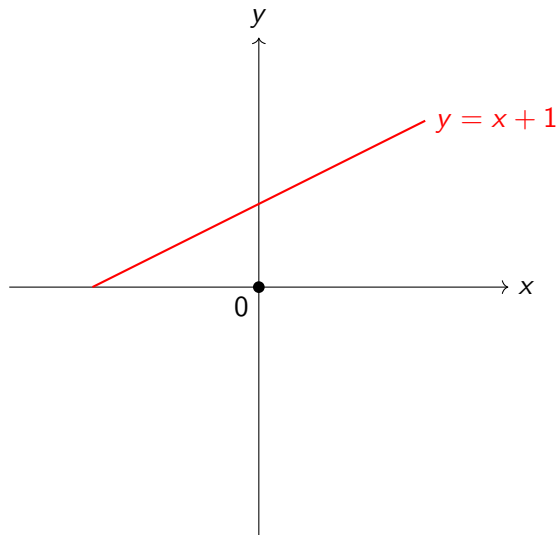
Subspace ✓

Subspace: Plane Through the Origin



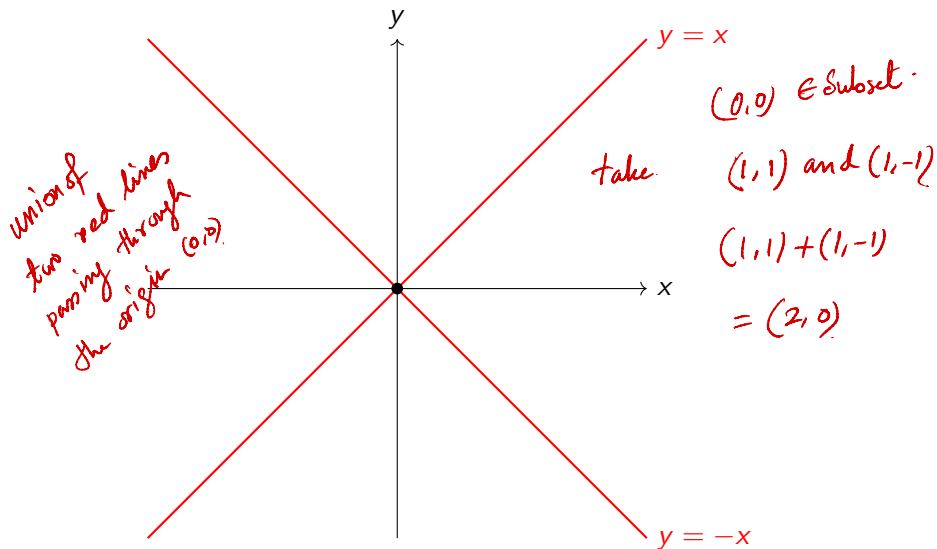
$$W = \{(x, y, 0) : x, y \in \mathbb{R}\}$$

Not a Subspace: Line Not Through Origin



Why not a subspace? $0 \notin W$.

Not a Subspace: Union of Two Lines



Why not a subspace? Not closed under addition.

Linear Combinations

Definition (Linear Combination)

Given $v_1, \dots, v_k \in V$, a vector of the form

$$\alpha_1 v_1 + \dots + \alpha_k v_k, \quad \alpha_i \in \mathbb{F} \quad (\mathbb{R})$$

is called a linear combination.

$$(2,0), (3,1), (1,-3) \in \mathbb{R}^2 \quad \leftarrow$$

$$1 \cdot (2,0) + 0.5(3,1) + 0.7(1,-3) \longleftrightarrow \text{lin comb}^n \text{ of}$$

$$-1 \cdot (2,0) + 0(3,1) + 0(1,-3) \longleftrightarrow \text{''}$$

Tab 1.

A	B	C	D	K
0.1	0.5	0.01	0.02	0.1

$$\frac{1}{2}(0.1, 0.5, 0.01, 0.02, 0.1) \in \mathbb{R}^5 \quad +$$

Tab 2.

A	B	C	D	K
0.2	0.2	0.1	0.01	0.02

$$\frac{1}{2}(0.2, 0.2, 0.1, 0.01, 0.02) \in \mathbb{R}^5$$

Span of Vectors

Definition (Span)

The **span** of $S = \{v_1, \dots, v_k\}$ is $\text{span}(S) = \left\{ \sum_{i=1}^k \alpha_i v_i : \alpha_i \in \mathbb{F} \right\}$.

- $\text{span}(S)$ is a subspace $v_1, v_2, \dots, v_k \in V$, $\{v_1, v_2, v_3, \dots, v_k\} = S$
- Smallest subspace containing S



$$S = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$$

$v_1 \qquad v_2$

$$\text{span}(S) = \left\{ \alpha_1 v_1 + \alpha_2 v_2 : \alpha_1, \alpha_2 \in \mathbb{R} \right\}$$

$$\boxed{\text{span}(S) \subseteq V}$$

$$\alpha_1 = 0, \alpha_2 = 1 \quad 0 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 1 \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\alpha_1 = 1, \alpha_2 = 1; \quad 1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \in \text{span}(S)$$

$$\boxed{\text{span}(S)}$$

Linear Dependence

Definition (Linearly Dependent)

Vectors $v_1, \dots, v_k$ are **linearly dependent** if

$$\alpha_1 v_1 + \dots + \alpha_k v_k = 0$$

for some scalars not all zero.

- **Interpretation:** At least one vector is redundant.

$\alpha_k \neq 0$

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_{k-1} v_{k-1} = -\alpha_k v_k$$

$\Rightarrow v_k = -\frac{\alpha_1}{\alpha_k} v_1 + \frac{\alpha_2}{\alpha_k} v_2 + \dots + (-1) \frac{\alpha_{k-1}}{\alpha_k} v_{k-1}$

$\beta_i = -\frac{\alpha_i}{\alpha_k} \Rightarrow v_k = \beta_1 v_1 + \beta_2 v_2 + \dots + \beta_{k-1} v_{k-1}$

$i \in \{1, \dots, k-1\}$

Linear Independence

Definition (Linear Independence)

Vectors $v_1, \dots, v_k$ are **linearly independent** if

$$\alpha_1 v_1 + \dots + \alpha_k v_k = 0 \Rightarrow \alpha_1 = \dots = \alpha_k = 0.$$

- **Interpretation:** No vector can be written as a combination of others.

Important Facts

- Any set containing 0 is linearly dependent
- Any set of more than n vectors in $\mathbb{R}^n$ is dependent

Basis of a Vector Space

Definition (Basis of a Vector Space)

A set $\{v_1, \dots, v_n\}$ is a **basis** of V if:

- it spans V
 - it is linearly independent
- Every vector in V has a *unique* representation in terms of a basis.

Examples of Bases

- Standard basis of $\mathbb{R}^n$
- $\{1, x, x^2, \dots, x^n\}$ for polynomials of degree $\leq n$

Dimension

Definition (Dimension)

The **dimension** of V , denoted $\dim V$, is the number of vectors in a basis of V .

Theorem

Any two bases of a finite-dimensional vector space have the same number of elements.

Examples

- $\dim(\mathbb{R}^n) = n$
- $\dim(\mathcal{P}_n) = n + 1$
- $\dim(\{0\}) = 0$

Subspaces and Dimension

Theorem

If W is a subspace of a finite-dimensional vector space V , then

$$\dim W \leq \dim V.$$

Equality holds if and only if $W = V$.

Thank You

Questions?